COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010/DATA STRUCTURES AND ALGORITHM	**SECTION:**	CPE21S1
DATE PERFORMED:	1/10/2026	**DATE SUBMITTED:**	4/10/2026
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ACTIVITY NO. 10 **Graphs**

1. PROCEDURE OUTPUT

ILO A:

```
Graph adjacency list
(start_vertex, end_vertex, weight):
(0, 2, 4) (0, 1, 2)
(1, 4, 3)
(2, 3, 2)
(3, 1, 4)
(4, 3, 3)
```

ILO B:

```
1:      {2: 0}, {5: 0},
2:      {1: 0}, {5: 0}, {4: 0},
3:      {4: 0}, {7: 0},
4:      {2: 0}, {3: 0}, {5: 0}, {6: 0}, {8: 0},
5:      {1: 0}, {2: 0}, {4: 0}, {8: 0},
6:      {4: 0}, {7: 0}, {8: 0},
7:      {3: 0}, {6: 0},
8:      {4: 0}, {5: 0}, {6: 0},

DFS order of vertices:
1
5
8
6
7
3
4
2
```

B.1

```
1:      {2: 2}, {5: 3},
2:      {1: 2}, {5: 5}, {4: 1},
3:      {4: 2}, {7: 3},
4:      {2: 1}, {3: 2}, {5: 2}, {6: 4}, {8: 5},
5:      {1: 3}, {2: 5}, {4: 2}, {8: 3},
6:      {4: 4}, {7: 4}, {8: 1},
7:      {3: 3}, {6: 4},
8:      {4: 5}, {5: 3}, {6: 1},

BFS order of vertices:
1
2
5
4
8
3
6
7
```

The graph is undirected, but each connection is stored twice. BFS ignores the weights and visits vertices level by level. it is less efficient because some vertices enter the queue more than once.

B.2

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

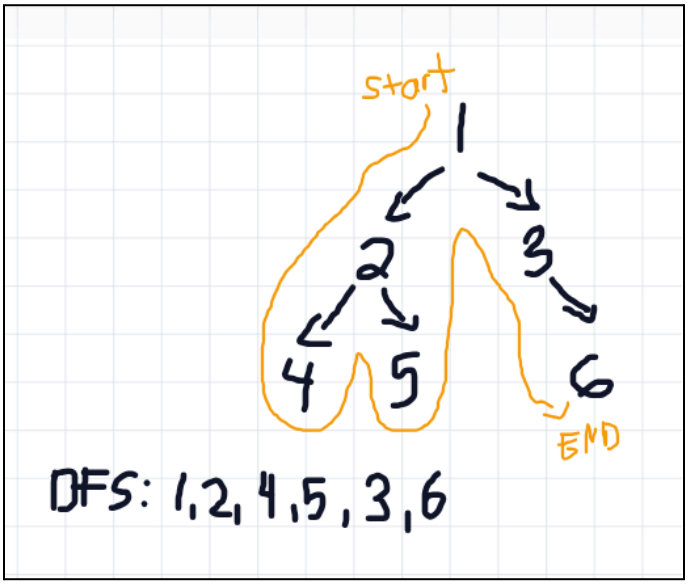
ILO C: Demonstrate an understanding of graph implementation, operations and traversal methods.

1. A person wants to visit different locations indicated on a map. He starts from one location (vertex) and wants to visit every vertex until it finishes from one vertex, backtracks, and then explore other vertex from same vertex. Discuss which algorithm would be most helpful to accomplish this task.

Depth-First Search (DFS) matches the situation because the person explores one path first, then backtracks when needed.

2. Identify the equivalent of DFS in traversal strategies for trees. To efficiently answer this question, provide a graphical comparison, examine pseudocode and code implementation.

Preorder is the closest to DFS because it visits a node first before going to its children.


start
1
2
3
4
5
6
END
DFS: 1,2,4,5,3,6

```cpp
#include <iostream>

struct Node{
    int val;
    Node* left;
    Node* right;
};
void preorder(Node* n){
    if (n == nullptr) return;
    std::cout<< n->val<<" ";
    preorder(n->left);
    preorder(n->right);
}

int main(){
    Node n4{4, nullptr, nullptr};
    Node n5{5, nullptr, nullptr};
    Node n6{6, nullptr, nullptr};
    Node n2{2, &n4, &n5};
    Node n3{3, nullptr, &n6};
    Node n1{1, &n2, &n3};

    preorder(&n1);
    return 0;
}
```

```
Problems  Output  Debug Console  Terminal  Po

PS C:\Users\loreen\Documents\CPE21S1\CPE0
sDebugLauncher.exe' '--stdin=Microsoft-MI
IEngine-Pid-0i2upmab.uiz' '--dbgExe=C:\ms
1 2 4 5 3 6
```

3. In the performed code, what data structure is used to implement the Breadth First Search?

BFS used a queue, which follows first-in first-out process. This visits vertices level by level. It also uses a set to track visited vertices and a vector to store the visit order.

4. How many times can a node be visited in the BFS?

Each node is visited only once because the 'visited' set keeps track of nodes that were already processed.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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